

Evaluation: Potential CLL-Classifiers

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1 Summary

Exp01

- Computed the RMA for the concatenated GSE13159 and GSE16455 superset (before I split train/ test)
- This is clear data leakage, because I transform train and test sets into a similar space, therefore test validations are invalid
- This is not how real-world ML prediction work... You usually take parameters from a train set to scale the test data
- This approach might be appropriate for biological processing, since it inherently removes noise

Exp02

- Classical Standard Scaling considering train and test set separate
- Meaning that I compute the test scaling from estimated parameters of the train set

Exp03

- Classical Robust Scaling considering train and test set separate
- Meaning that I compute the test scaling from estimated parameters of the train set

Exp04

- I got idea to approximate the RMA test sets with a shift-scale-vector estimated from train data

$$\begin{aligned}\mu^* &= \mu^{[1]} - \mu^{[0]} \\ \sigma^* &= \frac{\sigma^{[1]}}{\sigma^{[0]} + \epsilon} \\ \implies X^{scaled} &= (X^{unscaled} + \mu^*) * \sigma^*\end{aligned}\quad \text{Approximation}$$

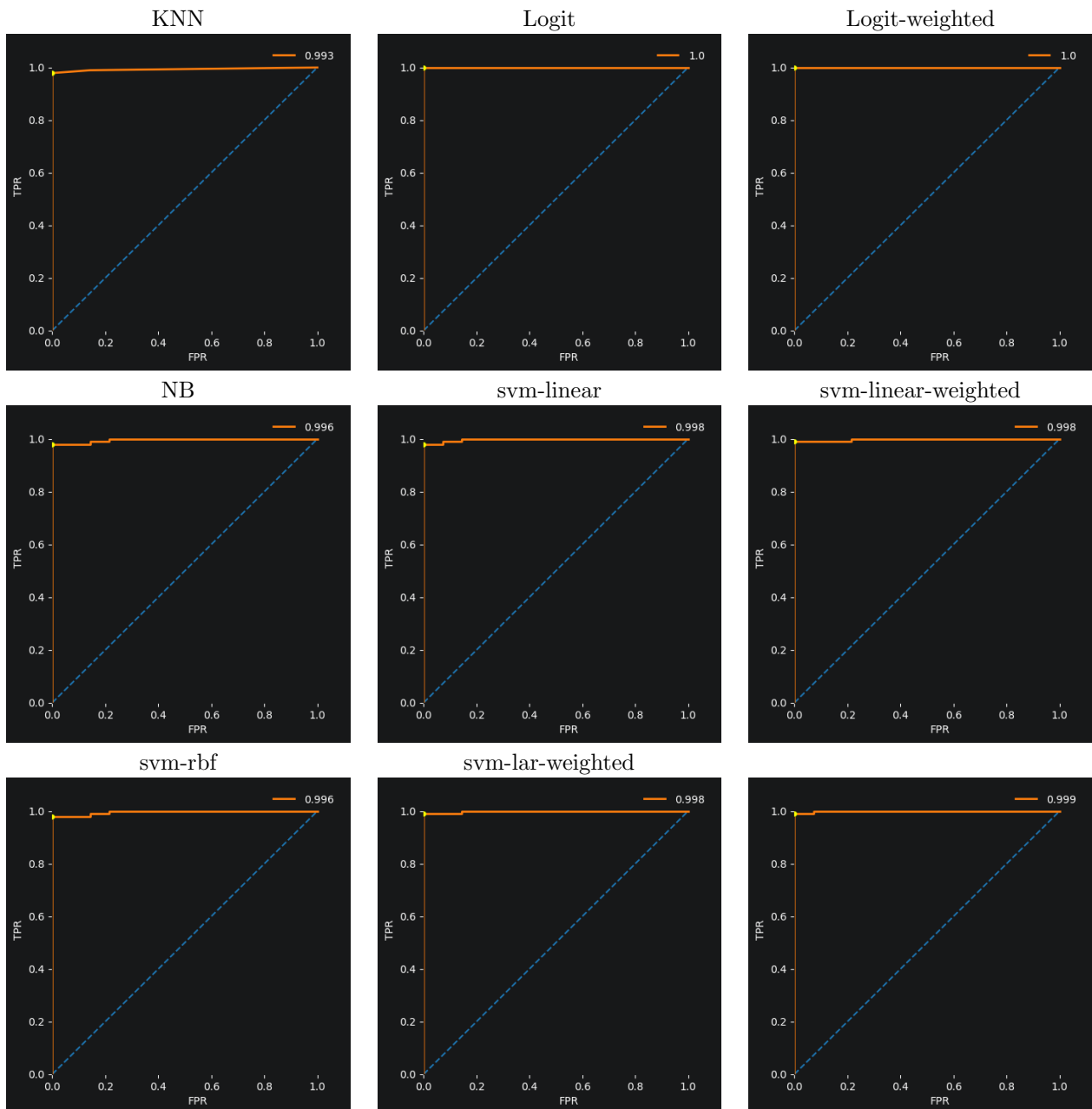
1.1 Conclusion

- The most valid models seems to be a class weighted linear SVM model onto the approximated scaling
- I aim that this model didn't learn experimental artifact but rather true biological signals, further validated cross-experimentally

2 Exp 01: Full Batch RMA

Since I introduced here intentionally strong data leakage, I will not investigate further after inspecting ROC curves of the trained model onto GSE131599-test.

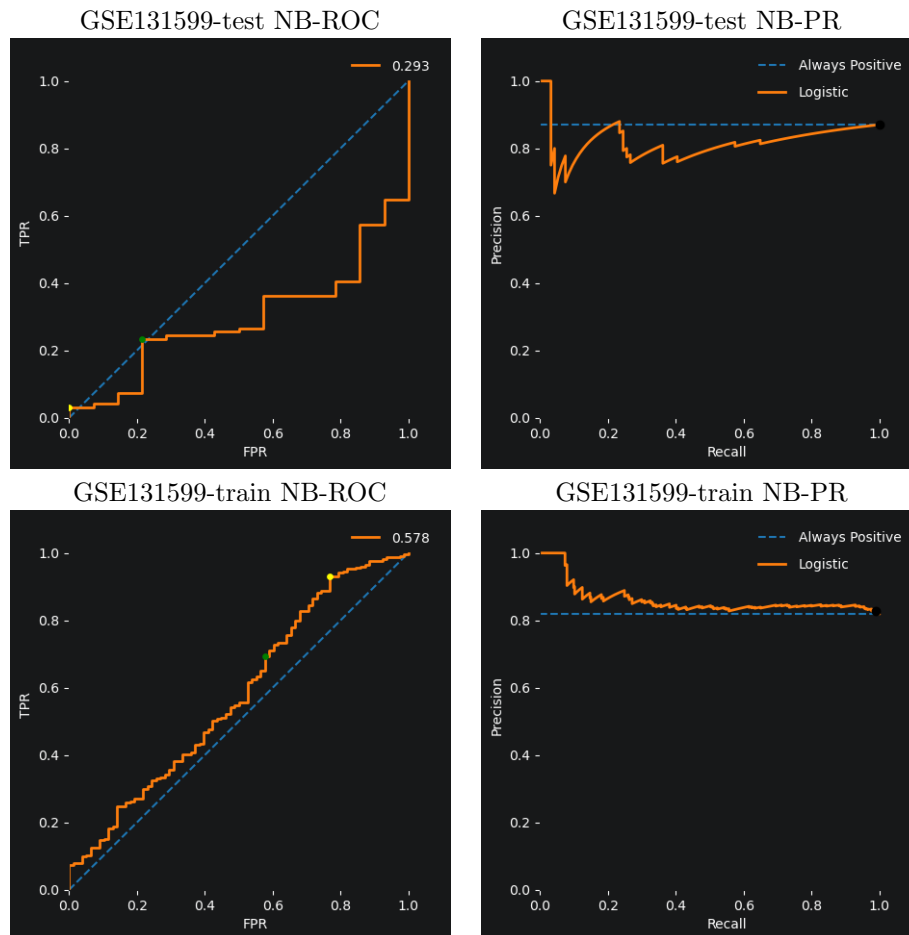
GSE131599-test (ROC)



- I intentionally produce some data leakage, I wanted to see the effect for scaling experiments combined with RMA
- This approach is clearly not valid, since it produces over-promising results
- Big Red Flag !
- Furthermore, also the prediction of GSE16455-test yields near perfect results, but i will not add those plots this report
- I will not further investigate here

3 Exp 02: Standard Scaling

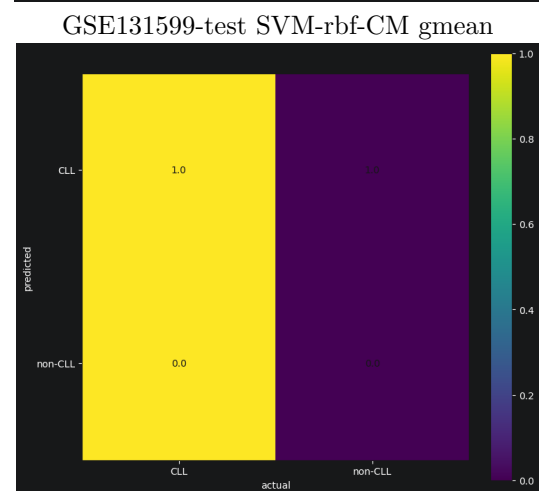
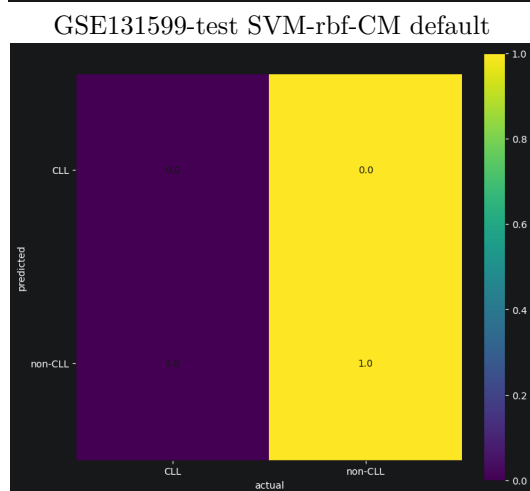
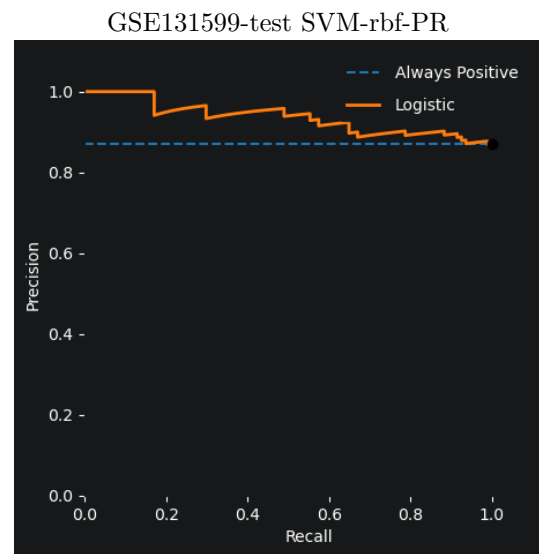
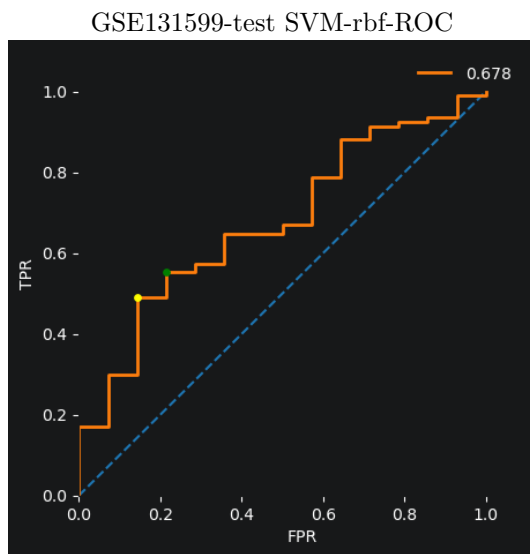
Most of the models ROC-AUC of the GSE131599-test data set was mostly around 0.5 and is therefore not a valid model for a diagnostic classifier. I only found that the Naive Bayes model reached a ROC-AUC of 0.293. I investigated further, since one could flip the prediction logic here, to get a ROC-AUC of roughly 0.7.



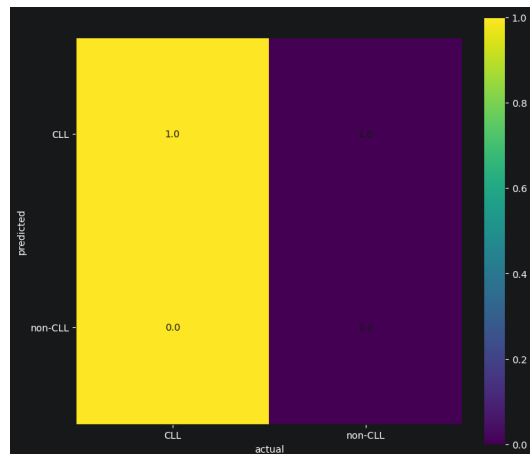
- Flipping the prediction logic, would be cherry-picking on the test data
- Also, it would make the train evaluation badly ...
- Conclusion, Standard Scaling with a Naive Bayes doesn't yield a promising classifier

4 Exp 03: Robust Scaling

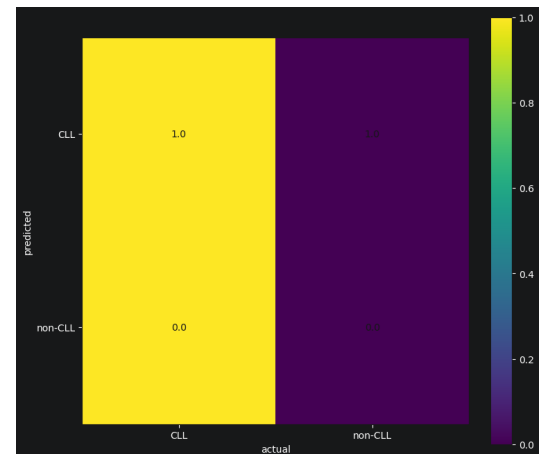
I found that the SVM-rbf performed onto the GSE13159-test very reasonable, so I investigated its performance further. I included the inspection of the confusion matrix (CM), when using different optimal threshold evaluated onto the train.



GSE131599-test SVM-rbf-CM Youden's J statistic



GSE131599-test SVM-rbf-CM F1-Score

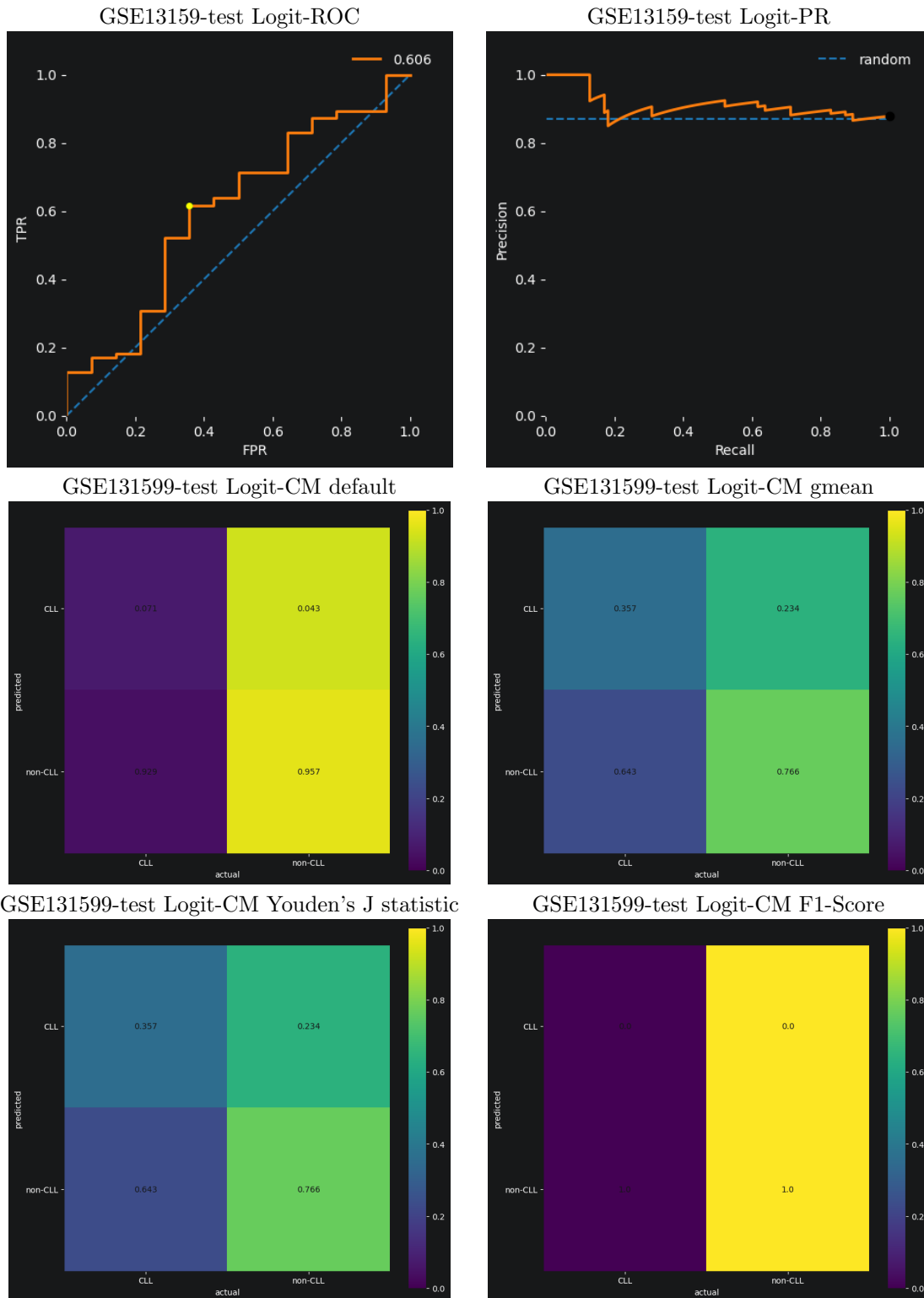


- One can clearly reason, that the ROC and PRC are promising, but looking at the actual predictions, the model only outputs a single class. that's a big Red flag, since the model doesn't generalize well (even with default threshold)

5 Exp 04: RMA-train with Approximation-test

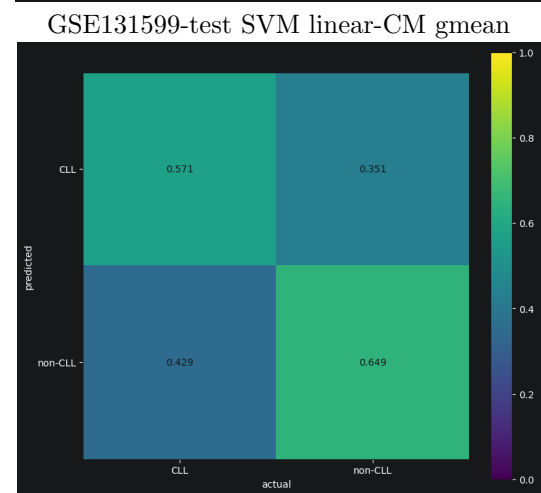
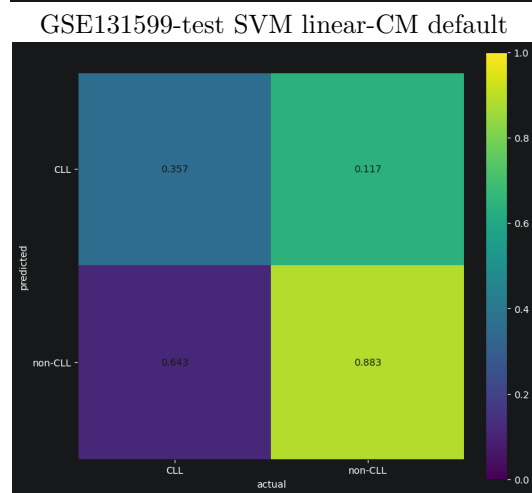
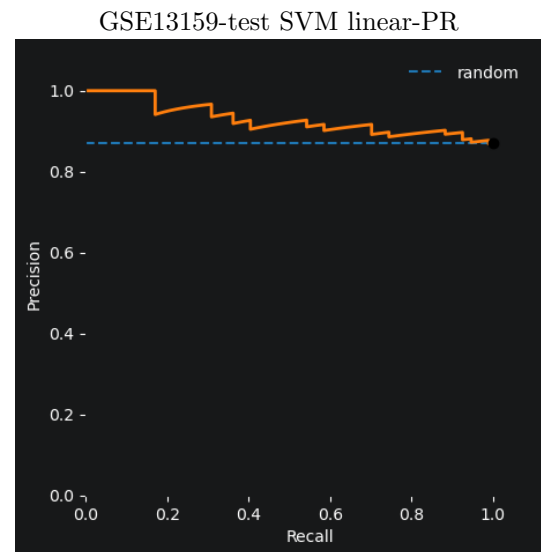
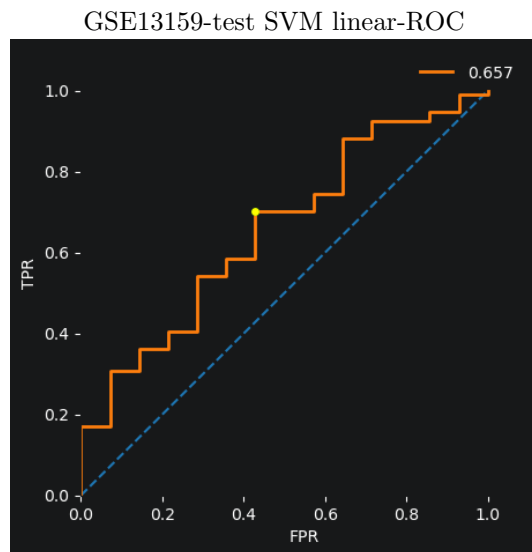
I found that 3 models (Logit, SVM-linear, SVM-linear-weighted) had an ROC-AUC > 0.6. Of course, this might not be the best performance, but considering that I approximated the RPM for the test data with a shift-scale-vector, working on noisy biological, older technology (microarray) and without batch correction, I would argue that this is the right direction to go. Those models could be a classifiers that actually learned biological signals and not just experimental artifacts ...

- Logit

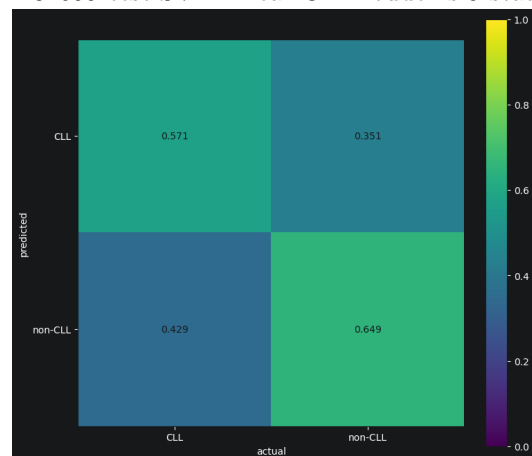


- The CM reveals that the model predicts non-CLL class okish, but performance undesirably, predicting the CLL classes, what is for an diagnostic classifier not appropriate

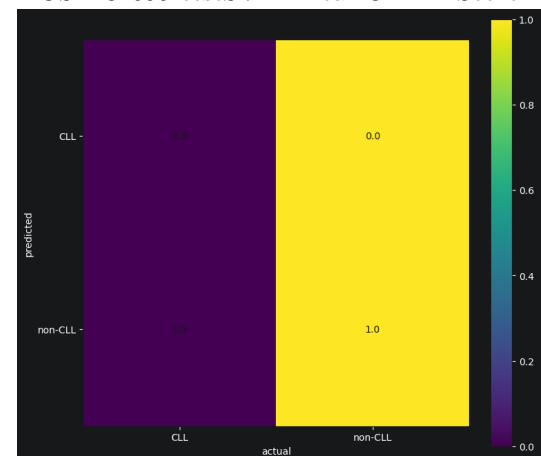
- SVM linear



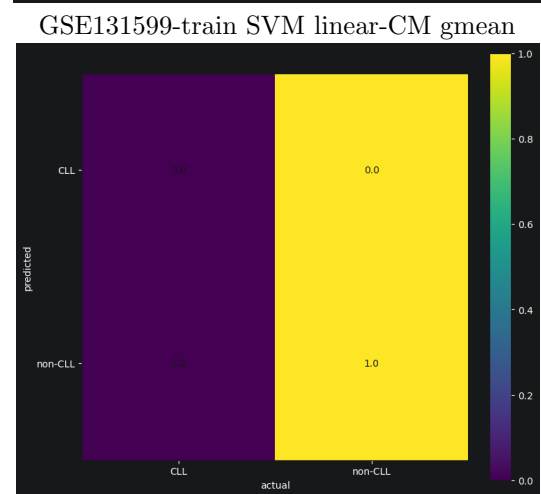
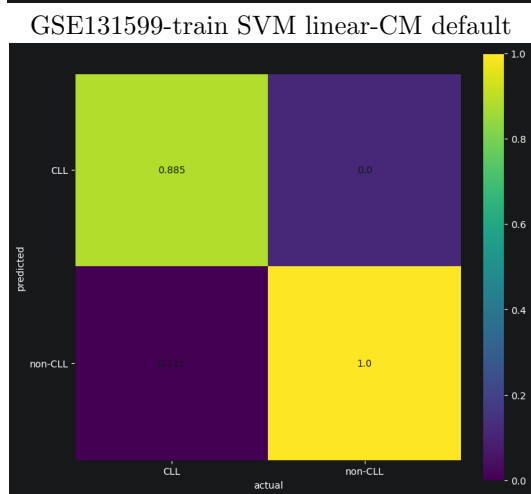
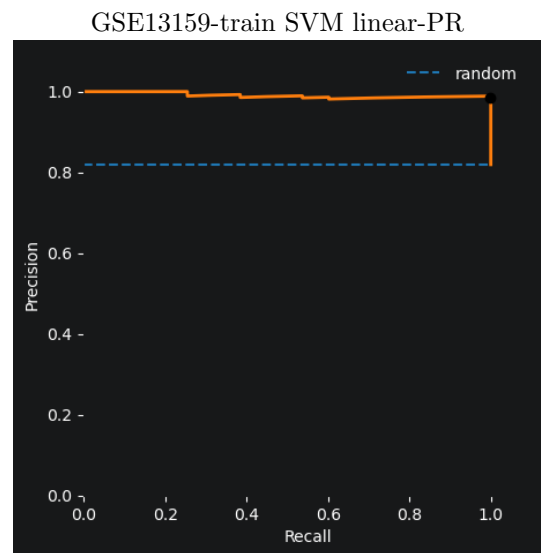
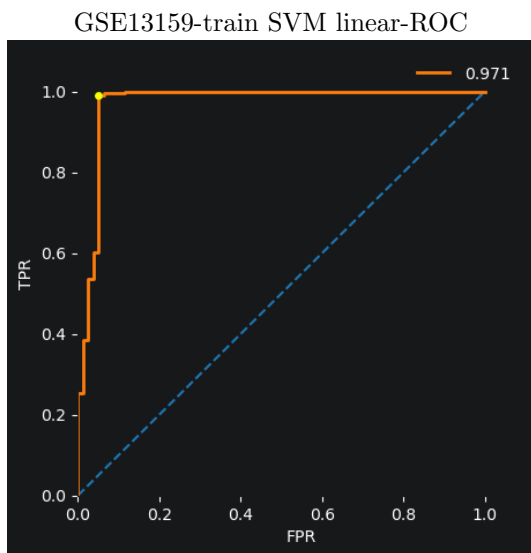
GSE131599-test SVM linear-CM Youden's J statistic



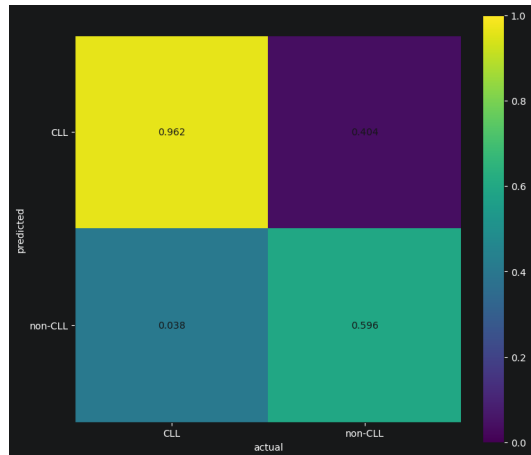
GSE131599-test SVM linear-CM F1-Score



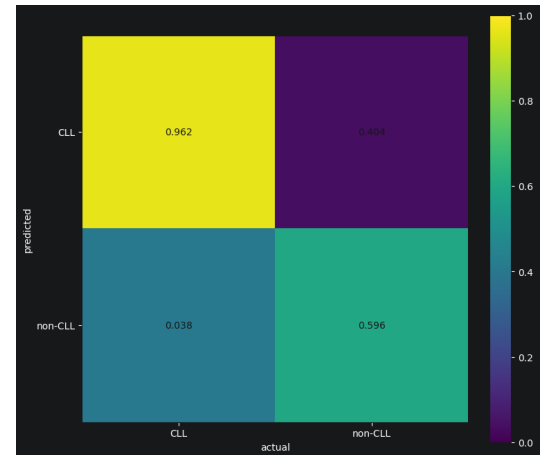
- The default threshold yield a very good non-CLL classifier, but doesn't perform well on the CLL classification
- Two threshold yields a kind of good direction, with 57.1% CLL correct and 64.9% non-CLL correct
- But i would sanity check the train ROC and the CM for the both threshold next



GSE131599-train SVM linear-CM Youden's J statistic

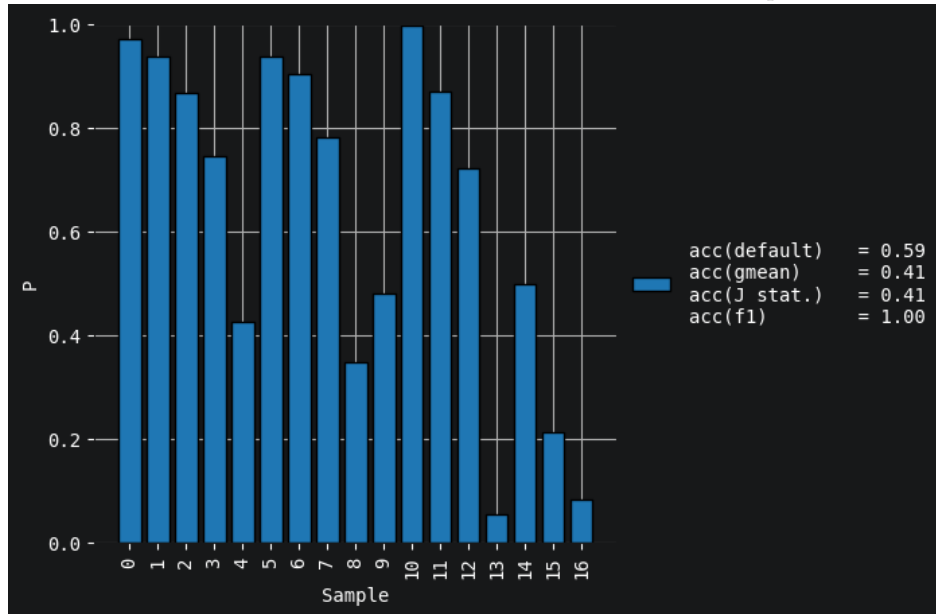


GSE131599-train SVM linear-CM F1-Score



- The classifier performs perfect onto the train data (data seems to be separable with an hyperplane)
- Using the gmean threshold, yield a classifier that only predicts non-cll, so not valid at all
- But using the Youden's statistic threshold, yield a reasonable training classification, and would align with the performance of the GSE13159-test
- So this model and this scaling seems to produce trustworthy predictions
- I would next inspect the GSE16355-test performance,

GSE16455-test SVM-linear Raw-Probabilities for samples

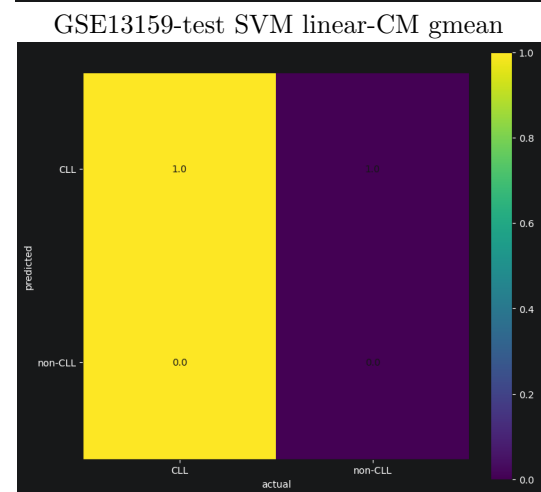
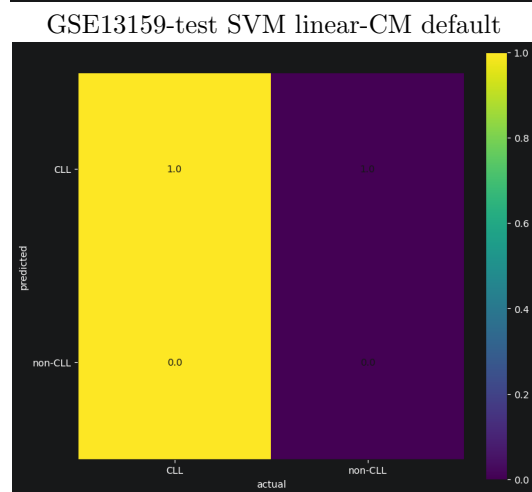
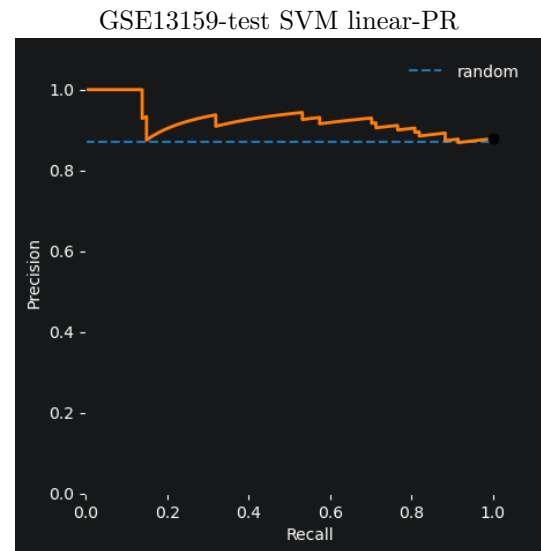
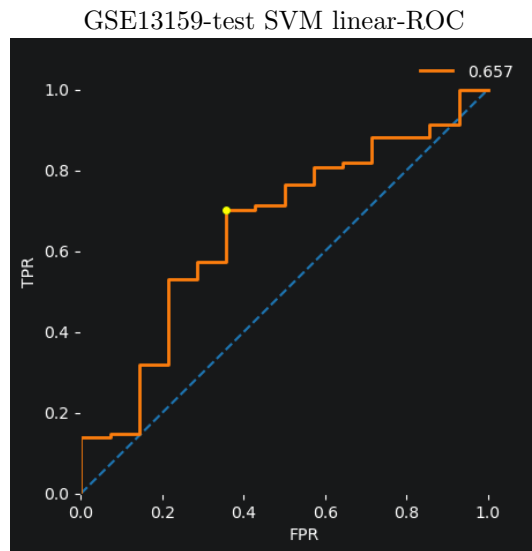


- GSE16455-test with $\text{acc}(\text{default}) = 0.59$ is fine, but model has very low true positives on the GSE13159-test (not valid)
- GSE16455-test with $\text{acc}(\text{gmean}) = 0.41$, yield a okisch classification for GSE13159-test, but only predicts non-CLL onto GSE13159-train (not valid)
- GSE16455-test with $\text{acc}(\text{J stat.}) = 0.41$, yield a okisch classification for GSE13159-test, also learned pattern in train data (most valid)
- GSE16455-test with $\text{acc}(\text{f1}) = 1.0$, only outputs non-CLL classification for GSE13159-test

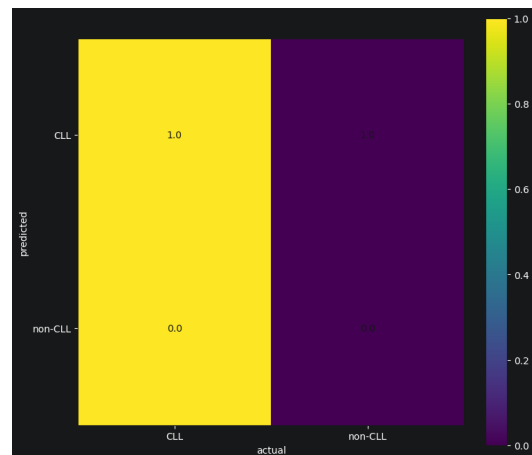
Conclusion

- I would argue that the SVM-linear onto the approxiamted scaling truely learns biological signals from this inherently noise data
- I can see that the shift-scaled approxiamtion of the RMA seems to work and performs better then just a standard or robust or standard scaling
- Considering the GSE13159-train, GSE13159-test and GSE16455-test evaluation, it seems that the optimal threshold for the Yourdan's statistics performs best

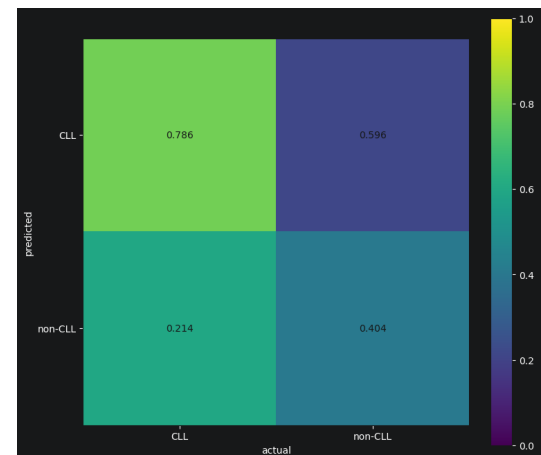
- SVM linear weighted



GSE131599-test SVM linear-CM Youden's J statistic

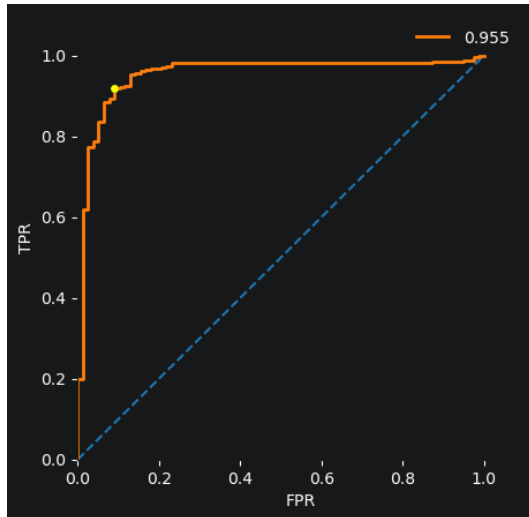


GSE131599-test SVM linear-CM F1-Score

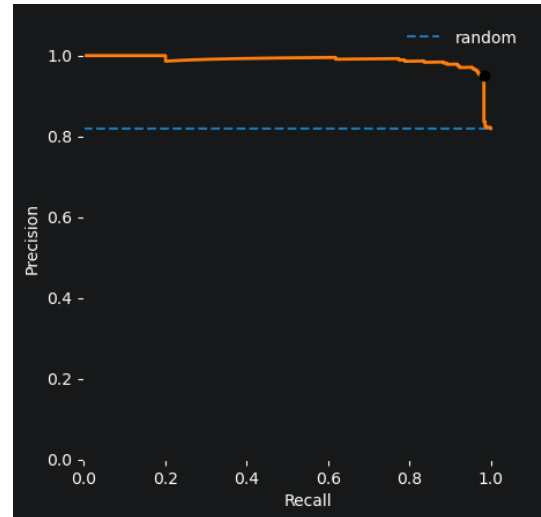


- The CM reveals that the model predicts reasonable with the f1-score optimizing threshold, the rest only output CLL classes
- I noticed that the correct classification of CLL classes is close to 0.8 but only roughly 0.4 non-CLL classes were labeled correctly
- Considering these numbers, this would be a better medical classifier, then the SVM-linear without class weights, because detecting CLL classes correctly, might be more relevant (diagnostically speaking), then non-CLL classes

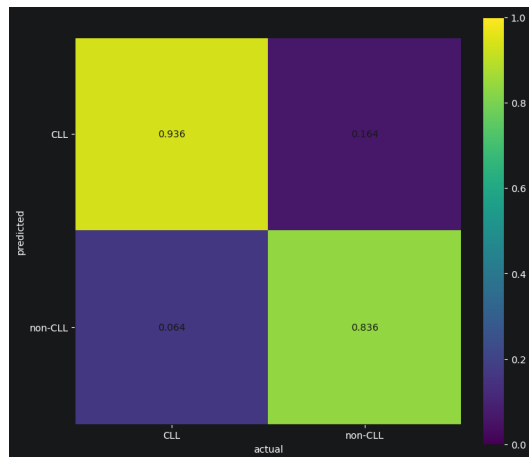
GSE13159-train SVM linear-ROC



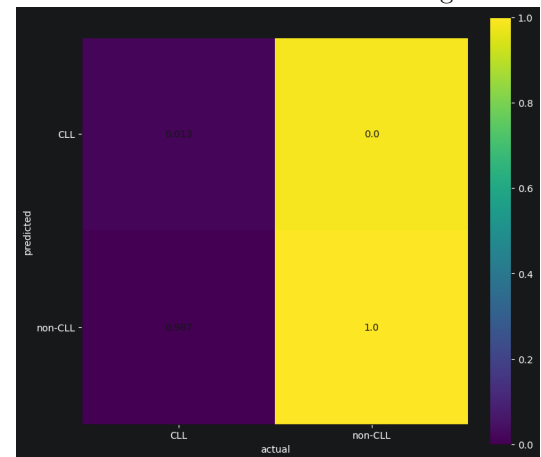
GSE13159-train SVM linear-PR



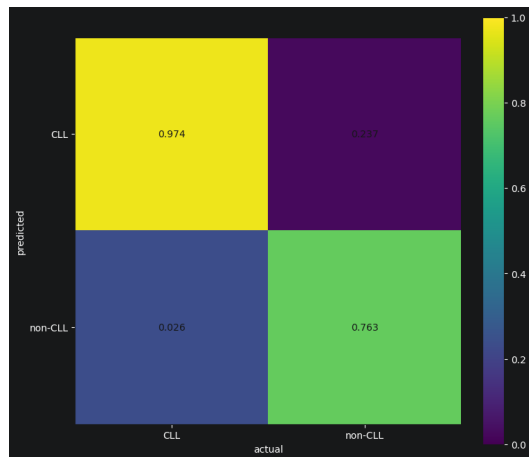
GSE131599-train SVM linear-CM default



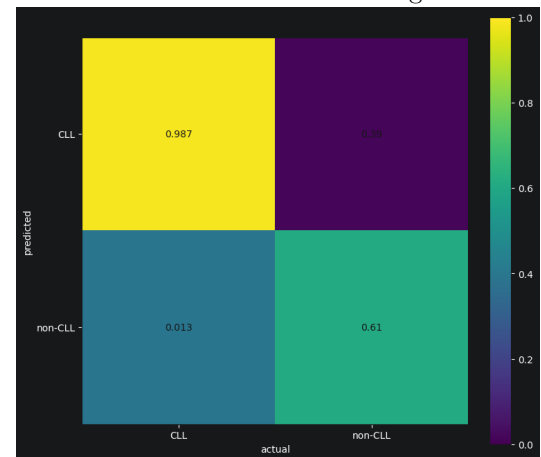
GSE131599-train SVM linear-CM gmean



GSE131599-train SVM linear-CM Youden's J statistic

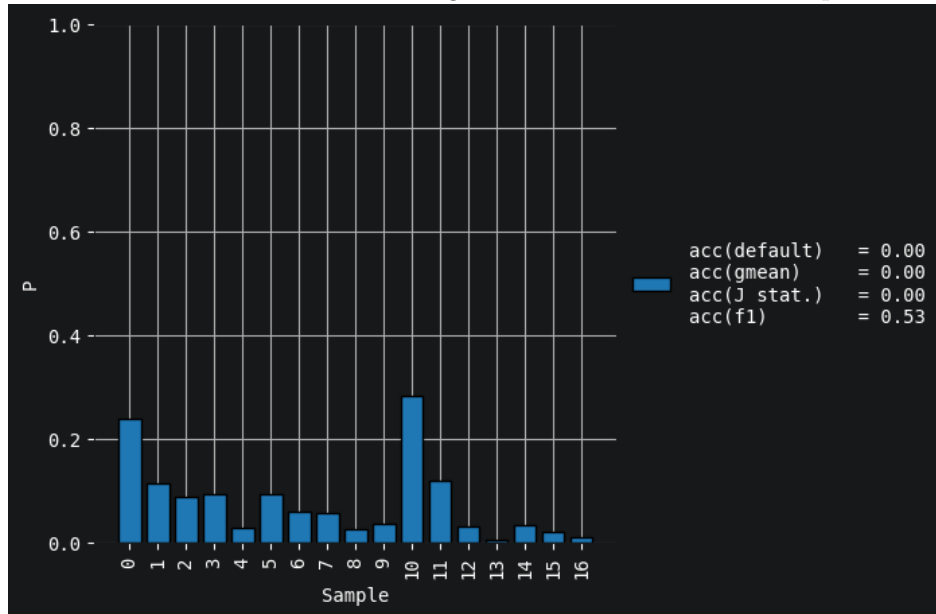


GSE131599-train SVM linear-CM-weighted F1-Score



- The CM reveals that the model predicts reasonable with the f1-score optimizing threshold for the same threshold then the GSE13159-test

GSE16455-test SVM-linear-weighted Raw-Probabilities for samples



- GSE16455-test with $\text{acc}(\text{default}) = 0.0$ (not valid)
- GSE16455-test with $\text{acc}(\text{gmean}) = 0.0$, (not valid)
- GSE16455-test with $\text{acc}(\text{J stat.}) = 0.0$ (not valid)
- GSE16455-test with $\text{acc}(\text{f1}) = 0.53$, this seems to be the most valid and promising model for this task, $CM_{\text{train}}(\text{CLL}) = 0.987$; $CM_{\text{train}}(\text{non-CLL}) = 0.61$; $CM_{\text{test}}(\text{CLL}) = 0.786$; $CM_{\text{test}}(\text{non-CLL}) = 0.404$

Conclusion

- The SVM-linear with class weights seem to be the most valid model
- Especially the fact, that for an medical classifier, the TP are high, without cherry picking for threshold, just using an mathematical optimal threshold